TutorTrace: Classifying Learner States in AI-Assisted Programming

UIST 2026 -- Session "Learning & Tutoring"

David Barron -- PRIME Lab, Virginia Tech

joint work with Minsun, Xiaohang, Rezky, Yue and Yan Chen

October 7, 2026

A student, 14 minutes into a lab

• Runs the code. IndexError.

• Asks the assistant. Pastes the fix. Runs. IndexError.

• Asks again, same words, plus "still".

• Stops typing for four minutes.

Is this student learning, or stuck?

Instructors can't see the process

- Autograders see submissions, not sessions
- Assistants see prompts, not what happens in the editor
- TAs see the 5 students who raise their hands

TutorTrace

- Dataset: replayable sessions combining editor events, runs and assistant prompts
- Taxonomy: six learner states, labeled in 30-second windows
- Classifier: predicts states from telemetry

Data: a live CS2 deployment

• CS 3114, Fall 2025: 87 students, 2 lab sections, 3 TAs

• Instrumented editor + assistant panel, per-stream consent

• 2,340 thirty-second windows labeled by 3 raters

• One section lost events to a reconnect bug – windows excluded, list released

Six states

- Productive progress
- Exploratory tinkering
- Stalled after error
- Assistant help-seeking
- Verifying (run & inspect)
- Disengaged

Stalled-after-error was added in labeling guide v3.

Reliability

	Fleiss' kappa
Overall (3 raters, 2,340 windows)	0.71
Assistant help-seeking	0.84
Tinkering vs. stalled-after-error	0.58

What predicts stalling

- Repeated near-identical prompts to the assistant
- Long idle gap after a failed run
- Reverting the assistant's change

Consistent with Ostrowski et al., L@S 2025.

Classifier

- Gradient boosting over keystroke, run and prompt features
- Splits grouped by student
- Macro-F1 0.78 in-course

Does it transfer? Cross-course analysis

- Second course: intro Python, Spring 2026
- Train on CS 3114, test on the second course
- Macro-F1 0.64 raw, 0.69 with per-course normalization
- Stalled-after-error transfers worst: it depends on the error channel

Design implications

- Flags as prompts for a human check-in, not automatic interventions
- Replay as a second stage for flagged sessions
- Consent per data stream; prompts are the most sensitive

Limitations

- Two courses, one institution
- Labels are judgments, even at kappa = 0.71
- States depend on the assistant students used

Thank you

Dataset, labeling guide and replay viewer:

(link on the paper page)

David Barron dbarron410@vt.edu